

29 Introduction to organic chemistry

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

At A Level, organic pathways integrate reaction mechanisms, stereochemistry and aromaticity. Optical isomerism occurs when a tetrahedral carbon has four different groups; a racemate has equal enantiomers and no net optical rotation. Mechanisms must account for electron movement and the stability of intermediates.

Worked example: follow the reasoning

Reduction of an unsymmetrical planar ketone can attack from either face, producing equal enantiomers if a new stereogenic centre results under achiral conditions.

Quick check · multiple choice

A Why is a racemate optically inactive?

[1]

A Equal opposite rotations cancel

B No chiral molecules exist

C Molecules are all planar

D It cannot absorb light

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Why is a racemic mixture optically inactive even if each pure enantiomer rotates plane-polarised light?

[2]

Answers and explanation

Quick check: A. Equal opposite rotations cancel

Question 1 [2 marks]

The mixture has equal amounts of opposite enantiomers [1]; rotations are equal and opposite so cancel [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

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